MEIL ESG & BRSR Platform

Full frontend, backend, architecture, workflow, RBAC, user capability, and implementation report

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Data classification: illustrative/demo environment; password hashes and credentials intentionally excluded.

Executive summary. This is a full-stack Next.js App Router application for multi-entity ESG and SEBI BRSR data governance. It combines role-specific React dashboards, server-side API routes, Prisma ORM over SQLite, deterministic validation/calculation helpers, evidence management, multi-level approvals, BRSR mappings, audit history, and PDF/report generation.

1. System at a glance

Area	Implementation-backed detail
Product	MEIL ESG & BRSR Core reporting and disclosure platform
Frontend	React 19, Next.js 16.1.1 App Router, TypeScript 5, Tailwind CSS 4, Radix UI/shadcn-style primitives, Recharts, Lucide, Framer Motion, Zustand/React Query utilities
Backend	Next.js route handlers under src/app/api, Prisma 6.11.1 client, SQLite database, bcryptjs password verification, signed HttpOnly cookie session, Zod validators
Reporting	Client-side jsPDF/autotable dependency plus server-side ReportLab/Python dashboard PDF generator and BRSR PDF generator
Deployment	Next standalone output; npm/Bun scripts; Caddy reverse-proxy configuration; Python 3.10+ for ReportLab rendering
Current runtime snapshot	15 users, 4 projects, 1 submissions, 4 evidence records, 7 audit logs, 5 activities

2. Architecture and system boundaries

Logical flow

Browser UI → AppProvider/session bootstrap → role-specific dashboard/workspace → fetch to Next.js route handler → server session/RBAC check → Prisma query or domain helper → SQLite → audit/activity/history/notification side effects → JSON/PDF response → dashboard/report view

Layer	Responsibilities	Representative files
Presentation	Welcome screen, AppShell, role dashboards, workspaces, reusable UI primitives, charts, dialogs, forms, responsive styling	src/app/page.tsx; src/components/dashboard; src/components/modules; src/components/ui; src/app/globals.css
Client state	Current user, active module, role-aware navigation, submodule routing, toast/error boundary behavior	src/lib/auth-context.tsx; src/lib/role-nav.ts; src/components/shell
HTTP/API	Authentication, ESG source records, evidence, submissions, validation, calculations, consolidation, BRSR, reports, audit, notifications	src/app/api/**/route.ts
Domain services	Workflow transitions, audit appenders, activity/history/notifications, contributor and finance stores, BRSR resolution, level-record fallback	src/lib/workflow.ts; src/lib/validators.ts; src/lib/brsr-resolver.ts; src/lib/contributor-store.ts; src/lib/finance-store.ts
Persistence	Relational organization hierarchy, RBAC, periods, source records, evidence, validation, calculations, submissions, BRSR, reports, audit	prisma/schema.prisma; src/lib/db.ts; db/custom.db
External/runtime boundary	Standalone Next server; Caddy reverse proxy; local file evidence paths; Python ReportLab process for dashboard export	next.config.ts; Caddyfile; scripts/generate_dashboard_pdf.py

Organization hierarchy

The relational hierarchy is Group → Subsidiary → Business Unit → Project/Site. Users receive scope rows at GROUP, SUBSIDIARY, BUSINESS_UNIT, or PROJECT level. The current runtime data contains one MEIL group, two subsidiaries, three business units, and four projects.

Group	Subsidiary	Business unit	Project
MEIL Group	MEIL Power Systems	Power Generation	MEIL-SOL-GJT — Gayatri Solar Plant (Gayatri, Telangana)
MEIL Group	MEIL Power Systems	Power Generation	MEIL-SOL-NZR — Nizamabad Solar Farm (Nizamabad, Telangana)
MEIL Group	MEIL Power Systems	Transmission & Distribution	MEIL-TD-HYD — Hyderabad 33kV Substation (Hyderabad, Telangana)
MEIL Group	MEIL Infrastructure	Water Infrastructure	MEIL-WTR-KPR — Kaleshwaram Lift Irrigation (Jayashankar, Telangana)

3. Frontend details

• Next.js App Router entry point is src/app/page.tsx; the page is a client boundary that wraps AppProvider, WelcomeScreen, AppShell, ErrorBoundary, role dashboard selection, and ModuleRouter.

• Role selection is explicit: the role key selects one of 16 dashboard implementations, including Super Admin, project/site, finance, HR, EHS, procurement, CSR, compliance, three reviewer levels, ESG manager/analyst, BRSR manager, auditor, and executive.

• Navigation is role-specific in src/lib/role-nav.ts. Each role receives a curated set of modules rather than the full application menu; module keys route to specialized workspaces in src/components/modules/module-router.tsx.

• UI composition uses Tailwind utility classes, CSS variables, Radix primitives, custom components, charts via Recharts, icons via Lucide, animation via Framer Motion, and accessible dialog/form/table primitives.

• Client/server trust boundary is intentional: the frontend mirrors session state for rendering, but backend authorization is enforced by server helpers before protected operations.

Frontend capability	Implementation evidence
Dashboards	23 dashboard TSX files; role-specific KPI, task, trend, risk, assurance, and readiness views
Workspaces	29 module TSX files covering shared, functional, reviewer, BRSR, audit, finance, and executive workspaces
UI system	48 reusable UI primitive files plus components.json and Tailwind configuration
Interaction patterns	Command palette, drawers, dialogs, tabs, tables, pagination, cards, date/calendar controls, validation modals, CSV import, toast notifications
Data presentation	Overview KPIs, monthly trends, target tracking, pipeline/submission state, anomaly/exception panels, audit trace and report previews

4. Backend details

Backend concern	Observed implementation
Framework runtime	Next.js 16 App Router route handlers; next.config.ts uses output: standalone and does not ignore TypeScript build errors
Authentication	POST /api/auth/login verifies bcrypt hash and returns a 12-hour HttpOnly, SameSite=Lax cookie named meil_esg_session; logout clears it; /api/auth/me exposes the current session projection
Authorization	requireUser and requirePermission load roles/scopes server-side; userHasPermission resolves RolePermission → Permission. Route handlers return 401/403 through authErrorToStatus patterns
Validation	Zod is installed and domain validation is implemented in validators.ts; validation results carry rule code, severity, message, field, status, resolution, and timestamps
Persistence	Prisma Client against SQLite; schema has organization, RBAC, period, unit/factor, ESG record, evidence, validation, calculation, submission, BRSR, report, notification, activity, and audit models
Auditability	appendAudit, appendActivity, appendHistory, and notifyUser persist actor, role, action, entity, state changes, reasons, comments, metadata, and links
PDF/reporting	BRSR generation is in src/lib/brsr-pdf-generator.ts; dashboard export route and scripts/generate_dashboard_pdf.py provide PDF output; jsPDF and jspdf-autotable are client dependencies

5. Data model and control chain

Source-to-disclosure chain: Source Data → Evidence → Validation → Calculation → Submission → BU Review → Subsidiary Review → HQ Review/Lock → Consolidation → BRSR Mapping → Report → Audit

Domain	Primary models / records	Purpose
Identity/RBAC	User, Role, Permission, RolePermission, UserRole, UserScope	Identity, role catalog, permission grants, and data boundary scopes
Organization	Group, Subsidiary, BusinessUnit, Project	Multi-entity reporting hierarchy and project/site ownership
Periods	ReportingYear, ReportingPeriod	Monthly/quarterly/annual period deadlines and lifecycle status
Calculation	UnitMaster, ConversionRule, EmissionFactor, CalculationResult	Normalization and deterministic energy/emission/resource calculations with factor version lineage
ESG source	EnergyRecord, WaterRecord, WasteRecord, WorkforceRecord, SafetyRecord, TravelRecord	Operational environment, people, safety, and travel inputs
Controls	Evidence, ValidationRule, ValidationResult, AnomalyEvent	Document proof, data quality rules, exceptions, and anomaly state
Workflow	Submission, SubmissionStatusHistory, CorrectionRequest	Submission package, review assignment, correction cycle, approval and lock
Disclosure	BrsrFramework, BrsrSection, BrsrPrinciple, BrsrQuestion, BrsrAnswer, Report	BRSR framework content, mappings, readiness, and generated reports
Trace	AuditLog, Activity, Notification	Immutable-style action history, project activity feed, and user notifications

Runtime database inventory

SQLite table	Rows
Activity	5
AnomalyEvent	0
AuditLog	7
BrsrAnswer	21
BrsrFramework	1
BrsrPrinciple	9
BrsrQuestion	21
BrsrSection	3
BusinessUnit	3
CalculationResult	6
ConversionRule	6
CorrectionRequest	0
EmissionFactor	10
EnergyRecord	6
Evidence	4
Group	1
Notification	4
Permission	28
Project	4
Report	1
ReportingPeriod	3
ReportingYear	1
Role	15

SQLite table	Rows
RolePermission	99
SafetyRecord	1
Submission	1
SubmissionStatusHistory	3
Subsidiary	2
TravelRecord	0
UnitMaster	9
User	15
UserRole	15
UserScope	14
ValidationResult	0
ValidationRule	0
WasteRecord	3
WaterRecord	2
WorkforceRecord	2

The active .env points to db/custom.db. A second prisma/db/custom.db exists with a different snapshot. This is a material environment/data-consistency issue and should be resolved before production use.

6. Workflow and state machine

Step	State/action	Primary actor or control
1	Create source record in DRAFT	Project, Finance, HR, EHS, Procurement, CSR, or Compliance contributor
2	Upload/link evidence	Contributor uploads document metadata; reviewer can verify or reject evidence
3	Validate and calculate	Validation rules flag INFO/WARNING/ERROR/BLOCKING; calculation helpers normalize units and compute results
4	Submit package	Contributor submits module package; rollup stores completion, evidence count, passed checks, and errors
5	Review/correct	BU reviewer reviews; correction request can return the package to DRAFT/SUBMITTED/PENDING; resubmission creates history
6	Approve at BU	BU_REVIEWER advances to BU_APPROVED or rejects
7	Approve at subsidiary	SUBSIDIARY_REVIEWER advances to SUBSIDIARY_APPROVED
8	HQ review and lock	GROUP_REVIEWER advances through HQ_REVIEW to LOCKED; locked records are not editable through normal endpoints
9	Consolidate and disclose	Group/subsidiary rollups feed BRSR mappings, readiness, report generation, and audit trace

Allowed transitions implemented in workflow.ts include DRAFT → SUBMITTED/PENDING → UNDER_REVIEW → CORRECTION_REQUESTED → RESUBMITTED, and approval progression through BU_APPROVED → SUBSIDIARY_APPROVED → HQ_REVIEW → LOCKED. The implementation also supports APPROVED and ACTIVE compatibility states.

7. User inventory and work capability

The configured runtime database contains 15 active demo users. The seed defines a 16th FINANCE_USER (finance@meil-esg.in) that is present in prisma/db/custom.db but missing from db/custom.db. All listed identities are illustrative; no password or password hash is included.

User	Email	Employee code	Role	Scope	Work / ability
Anita Desai	anita@meil-esg.in	MEIL-ESG-001	ESG_MANAGER	GROUP: MEIL Group	Owns ESG KPI quality, completeness, material risks, targets, cross-functional review, and methodology oversight.
Arjun Mehta	admin@meil-esg.in	MEIL-ADM-001	SUPER_ADMIN	Group-wide / unrestricted	Administers users, roles, organization, projects, configuration, security, and all workflows.
Deepika Joshi	deepika@meil-esg.in	MEIL-GOV-005	COMPLIANCE_USER	GROUP: MEIL Group	Maintains policies, obligations, controls, cases, ethics, and the regulatory calendar.
Imran Sheikh	imran@meil-esg.in	MEIL-CSR-017	CSR_USER	SUBSIDIARY: MEIL Infrastructure	Maintains CSR projects, budgets, beneficiaries, impact, community engagement, and local sourcing data.
K. Venkat	kvenkat@meil-esg.in	MEIL-EHS-009	EHS_USER	PROJECT: Gayatri Solar Plant	Maintains safety incidents, inspections, corrective actions, environmental, energy, water, waste, and safety evidence.
Karthik Subramaniam	karthik@meil-esg.in	MEIL-AUD-001	AUDITOR	GROUP: MEIL Group	Performs assurance engagements, scope/materiality, evidence testing, BRSR testing, findings, requests, and reports.
Meena Iyer	meena@meil-esg.in	MEIL-BRSR-001	BRSR_MANAGER	GROUP: MEIL Group	Configures frameworks, manages Sections A/B/C and BRSR Core mappings, readiness, report building, and issuance.
Nisha Pillai	nisha@meil-esg.in	MEIL-SUB-001	SUBSIDIARY_REVIEWER	SUBSIDIARY: MEIL Power Systems	Reviews subsidiary consolidation, BRSR impact, evidence, and subsidiary approvals.
Priya Nair	priya@meil-esg.in	MEIL-PRC-031	PROCUREMENT_USER	SUBSIDIARY: MEIL Power Systems	Maintains suppliers, assessments, sustainable sourcing, transactions, and value-chain data.
Rajesh Khanna	rajesh@meil-esg.in	MEIL-EXE-001	EXECUTIVE	GROUP: MEIL Group	Consumes enterprise ESG, BRSR readiness, strategic risk, trends, business-unit, assurance, and report dashboards.
Rakesh Verma	rakesh@meil-esg.in	MEIL-BUR-002	BU_REVIEWER	BUSINESS_UNIT: Power Generation	Reviews business-unit submissions, evidence, exceptions, and performs BU approval or rejection.
Rohit Kumar	rohit@meil-esg.in	MEIL-PU-014	PROJECT_USER	PROJECT: Gayatri Solar Plant	Enters site/project ESG source data, uploads evidence, submits records, and tracks project reporting.

User	Email	Employee code	Role	Scope	Work / ability
Sameer Khan	sameer@meil-esg.in	MEIL-ANA-001	ESG_ANALYST	GROUP: MEIL Group	Explores ESG data, metrics, emissions, energy/resources, social/governance trends, variance, and quality.
Sunita Rao	sunita@meil-esg.in	MEIL-HR-022	HR_USER	SUBSIDIARY: MEIL Power Systems	Maintains workforce, training, wellbeing, human-rights, and workforce evidence data.
Vikram Shah	vikram@meil-esg.in	MEIL-GRP-001	GROUP_REVIEWER	GROUP: MEIL Group	Performs group consolidation, enterprise review, assurance review, final BRSR review, and locking.
Rakesh Verma	finance@meil-esg.in	MEIL-FIN-001	FINANCE_USER	Seed-defined; scope not present in runtime DB	Collects financial turnover, CapEx/OpEx, resource expenditure, supporting evidence, and submissions. (expected seed account; not in active runtime DB)

Role permissions

Role	Phase	Role purpose	Effective permission set
COMPLIANCE_USER	Phase 1	Governance & ethics	organization.read, project.read, evidence.read, submission.submit, brsr.read, report.read, audit.read
CSR_USER	Phase 1	CSR & community impact	organization.read, project.read, evidence.upload, evidence.read, submission.submit, brsr.read, report.read, audit.read
EHS_USER	Phase 1	Safety & environmental incidents	organization.read, project.read, esg.safety.write, esg.energy.write, esg.water.write, esg.waste.write, evidence.upload, evidence.read, submission.submit, brsr.read, report.read, audit.read
HR_USER	Phase 1	Workforce & HR data	organization.read, project.read, esg.people.write, evidence.upload, evidence.read, submission.submit, brsr.read, report.read, audit.read
PROCUREMENT_USER	Phase 1	Supplier & value chain	organization.read, project.read, esg.travel.write, evidence.upload, evidence.read, submission.submit, brsr.read, report.read, audit.read
PROJECT_USER	Phase 1	Project-level source data entry	organization.read, project.read, esg.energy.write, esg.water.write, esg.waste.write, esg.people.write, esg.safety.write, esg.travel.write, evidence.upload, evidence.read, submission.submit, brsr.read, report.read, audit.read
BU_REVIEWER	Phase 2	BU review & approval	organization.read, project.read, submission.review, submission.reject, submission.approve, evidence.read, evidence.verify, brsr.read, report.read, audit.read
GROUP_REVIEWER	Phase 2	Group final review & lock	organization.read, project.read, submission.review, submission.approve, submission.lock, evidence.read, brsr.read, brsr.generate, report.read, report.generate, audit.read

Role	Phase	Role purpose	Effective permission set
SUBSIDIARY_REVIEWER	Phase 2	Subsidiary consolidation review	organization.read, project.read, submission.review, submission.approve, evidence.read, brsr.read, report.read, audit.read
AUDITOR	Phase 3	Assurance read-only scope	organization.read, project.read, evidence.read, brsr.read, report.read, audit.read
BRSR_MANAGER	Phase 3	BRSR reporting workflow	organization.read, project.read, brsr.read, brsr.generate, brsr.configure, report.read, report.generate, audit.read
ESG_ANALYST	Phase 3	Analytics & trends	organization.read, project.read, brsr.read, report.read
ESG_MANAGER	Phase 3	ESG data quality & methodology	organization.read, project.read, evidence.read, brsr.read, report.read, audit.read
EXECUTIVE	Phase 3	Executive reporting access	organization.read, project.read, brsr.read, report.read
SUPER_ADMIN	Phase 3	Entire system administration	all 28 permissions

8. API surface

Area	Routes	Purpose
Auth	/api/auth/login, /logout, /me	Login, logout, session projection
ESG source	/api/energy, /water, /waste, /workforce, /safety, /data-entry	Create, update, list, and validate source data
Evidence	/api/evidence, /api/evidence/[id], /verify, /reject	Upload metadata, read, verify, reject
Workflow	/api/submissions and /api/submissions/[id]/submit|review|approve|reject|resubmit|lock|history	Submission lifecycle and status history
BRSR	/api/brsr/frameworks, sections, principles, questions, indicators, mappings, readiness, generate, preview	Framework, mapping, readiness, and document generation
Analytics	/api/overview, /insights, /targets, /consolidation/[level], /organization/tree	KPI, insights, target, consolidation, hierarchy data
Reports	/api/reports, /generate, /[id], /[id]/download, /export/dashboard-pdf	Report catalog, generation, download, dashboard PDF export
Governance	/api/audit, /audit/trace/[id], /activity, /notifications, /notifications/[id]/read	Traceability, activity feed, and notifications
Admin/controls	/api/calculation/run, /validation/run, /organization/tree	Calculation, validation, organization and control operations

9. Security and operational assessment

Control	Current implementation	Assessment
Password handling	bcryptjs compare against passwordHash	Good baseline for demo; use managed identity/rotation policy for production
Session	Signed token in HttpOnly cookie; 12-hour max age; SameSite=Lax	Server-side enforcement is present; token is lightweight and should be replaced with hardened secret rotation/session store in production
Secret safety	Production boot refuses missing SESSION_SECRET; fallback is explicitly demo-only	Good guardrail; ensure secret is set in deployment
RBAC	RolePermission checks are performed server-side by protected route handlers	Strong pattern; verify every route has scope enforcement, not only permission enforcement
Audit	State/action appenders persist actors, roles, old/new state, reasons and metadata	Good traceability model; enforce append-only database policy and retention in production
Evidence	Evidence stores file name, type, hash, MIME, version, verification status and reviewer	Good metadata model; file storage authorization and malware scanning should be production controls
Environment consistency	.env uses db/custom.db while prisma/db/custom.db contains a different seed snapshot	High-priority issue: standardize one database path and migration/seed procedure
Demo data	Seed marks users demo=true and documents illustrative data	Keep demo database isolated; rotate/remove shared demo credentials before deployment

10. Build, run, and deployment workflow

Stage	Command/configuration	Result
Install	bun install or npm install	Installs Next, React, Prisma, UI, chart, report, and validation dependencies
Database	bun run db:generate; bun run db:push	Generates Prisma client and applies schema to configured SQLite file
Development	bun run dev	Next dev with webpack on port 3000
Quality	npm run lint	ESLint over repository
Production build	npm run build	Next standalone build, copies static and public assets into standalone output
Production start	npm run start	Runs standalone server through Bun and writes server.log
PDF compiler	python scripts/generate_dashboard_pdf.py <json> <pdf>	ReportLab dashboard snapshot; this report uses the new generator below
Reverse proxy	Caddyfile	Routes public traffic to the standalone application according to deployment configuration

11. Findings and recommended next actions

• Resolve the database mismatch: the configured runtime db/custom.db has 15 users and 1 submission, while prisma/db/custom.db has 16 users and 18 submissions. Pick one canonical database path and remove ambiguity from scripts and deployment.

• Restore or intentionally omit the seed-defined Finance user in the runtime DB; the report marks finance@meil-esg.in as expected but absent from db/custom.db.

• Before production, set a strong SESSION_SECRET, remove shared demo credentials, use a real secret manager, and consider a database-backed session or established auth provider.

• Complete scope-level authorization checks consistently across every record query; permission checks alone do not prove that the user can access the requested project/subsidiary/group.

• Move evidence files to controlled object storage with signed URLs, malware scanning, content-type verification, encryption, retention policy, and access logging.

• Add automated integration tests for login/RBAC, each workflow transition, lock immutability, scope isolation, report generation, and database path selection.

• Use Prisma migrations rather than db push for controlled environments, and document backup/restore, SQLite concurrency limits, and the production database migration path.

Source files reviewed: package.json, README.md, next.config.ts, tsconfig.json, tailwind.config.ts, prisma/schema.prisma, prisma/seed.ts, src/app/page.tsx, src/lib/session.ts, src/lib/workflow.ts, src/lib/role-nav.ts, src/components/modules/module-router.tsx, src/app/api/**/route.ts, scripts/generate_dashboard_pdf.py, Caddyfile, and the configured SQLite runtime snapshot.